PANTRYFINDER: ONLINE COMMUNITY PANTRY MAPPING AND DONATIONS MANAGEMENT SYSTEM UTILIZING GPS TECHNOLOGY

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ABSTRACT

This study developed PantryFinder: Online Community Pantry Mapping and Donations Management System Utilizing GPS Technology is an online mobile application for citizens like pantry owners, pantry seekers and pantry donors. The application aimed to help the pantry seekers to easily find or locate different community pantries around their areas or any places they want as long as the pantry is available. For the pantry owner to easily create a post regarding their pantry so that everyone can see it and for the pantry donor to easily make a donation to the community pantry that lacks essential items.

The researchers use an agile method to make the development process efficient and be able to adapt changes based on its phases. The researcher evaluated the developed system using questionnaires to measure its level of acceptability, used weighted mean for the average and frequency, and convenience sampling for the data collection on every respondent. The answers were calculated and respondents rated the developed system as highly acceptable and was proven to be more efficient, functional and secure. It was concluded that the developed system will be a big help in terms of donations, finding and creating community pantries. Researchers also recommend to enhance the user interface, integrate advanced mapping, chat feature and must be available on every version of smartphones.

Keywords: *Community Pantry, Management, Donation, GPS*

1. INTRODUCTION

Nowadays, different disasters happened such as typhoons and pandemic like COVID-19 that makes people experience sufferings in life unexpectedly, they lose their precious jobs, unable to work to make money because of quarantine and that problems will cause them to starve and cannot provide the needs for themselves especially for their families. That's why community pantries are spreading around the world and the community pantry is not a new concept; many have been developed elsewhere as an urgent, local solution to hunger; but, in the light of the pandemic and millions of food-insecure Filipinos, the concept has proved nothing short of innovative [1]. There are some wholeheartedly kind persons who volunteered individually to donate some foods, clothes, hygiene kits or any essential items to help those people who are badly affected by every disaster.

A crowdsourced map was created by Filipino geographers that shows the different locations of all the country's community pantries. Saan May Community Pantry? is an interactive map which shows the locations of community pantries in the country represented by pins. Clicking on each pin will show the details of the community pantry. Andi Tabinas, a graphic designer and urban planner, and her team contributed "Saan May Community Pantry?" to put this into practice and bayanihan on show in the hundreds of community pantries that grow across the country. Before it became a community pantries map, it was "Saan May Nagdedeliver?" map, it was created to assist Filipinos to locate stores that provide delivery services when the country experienced the enhanced community quarantine (ECQ). However, since quarantine restrictions were relaxed and deliveries became more readily available, the map was not simply updated. They want it "to use the map for good" by the people and the organizers can request for the removal of their personal details on the map if they want for their privacy and safety. [2]

GPS has expanded into people's daily lives, where they can use it for navigation and for taking directions for travel. Furthermore, it is beneficial for a user to look for nearby facilities, particularly if they are unfamiliar with the new place. Locator Maps as a geographic guide are useful because they help to analyze and comprehend places and areas. Furthermore, mobile phones are technology that has been part of people's daily lives and Google maps on smartphones helps users in finding their destination in the most efficient way [3].

A geographic information system (GIS) is a computer system that captures, stores, verifies, and displays data relevant to locations on the Earth's surface. Despite the fact that entities and characteristics can be encoded in a variety of data files, with the use of GIS it will allow decision makers to classify locations or routes whose attributes meet various parameters. GIS can also help individuals and organizations better understand spatial structures and interactions by connecting apparently unrelated details. [4]

With the use of GIS and mapping systems, the process of searching and finding different places of community pantries will be easier for the user, also for the donors who want to donate some essential foods and goods.

The current method to know if there is a community pantry is to call and ask some people where they can find it, by checking on social media sites such as Facebook to see if there are posts about community pantry or by the information they can get from neighbors and friends. Even though hundreds of community pantries have sprouted across the different cities in the Philippines, some people still do not have any idea that there are community pantries around their area.

Therefore, researchers came up with the idea of helping people in finding different community pantries and monitor the status. They will know what the essential items are with that pantry, tracking the pantries that need a donation of essential items, and identify the number of donations that the community pantry received. This will also help the people in becoming more aware of social distancing because the system includes the protocols that they need to do.

2. OBJECTIVES

The general objective of this study is to develop a system for the people to have an easiest way of finding different community pantries within the area and donate some essentials.

The study also attempted to answer the following questions:

1. What are the problems encountered in the management of community pantries by the pantry owner, pantry donor, and pantry seeker?
2. What are the modules to be included on the developed system?
3. What are the software and hardware components needed to develop the developed system?
4. What are the technical evaluation results of the developed system?
5. What is the level of acceptability of the developed system in terms of:
    - 5.1 Functionality Suitability
    - 5.2 Performance Efficiency
    - 5.3 Usability
    - 5.4 Security

3. MATERIALS AND METHODS

Software Development

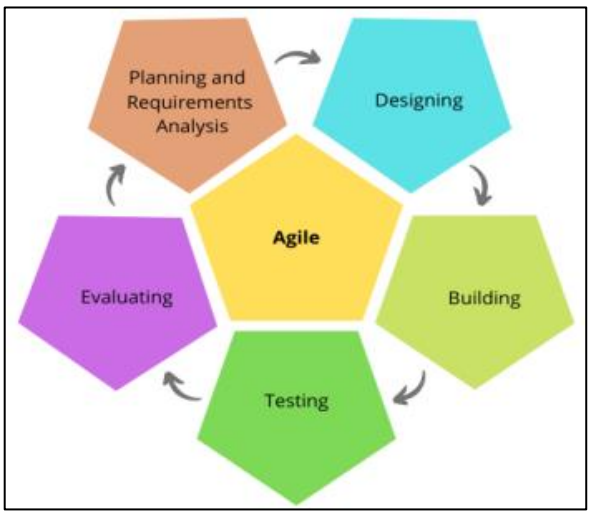

Figure 1. Agile development methodology

Figure 1 shows the process and methods in agile development methodology. It includes design phase, build phase, testing phase, evaluation phase, and planning and requirements analysis.

Planning and Requirements Analysis

In this phase, the researchers gathered information which talks about the developed system. They communicated with the stakeholders and the project team for the software development.

Hardware and Software Requirements

Table 1 and Table 2 are the hardware and software requirements needed to run the system.

Table 1. Hardware requirements

Particulars	Specification
Processor	2 GHz
Memory	4 GB
Storage	32 GB
Mobile Phones	Android Version

Table 1 shows the hardware needed to develop the developed system. The indicated hardware requirements such as Processor, Memory, Video Card and Mobile Phones are important to the developed system, as they were all needed to input the data of the developed system.

Table 2. Software requirements

Particulars	Software
Phone OS	Jelly Bean up to Nougat
Web Browser	Google Chrome

Table 2 shows the software requirements used to provide the functionality of the web system and application. A phone OS which is the Jelly Bean up to Nougat and web browsers like Google Chrome.

Designing Phase

In this phase, the team generated a variety of different outputs, including sketches, flowcharts and prototypes of the developed system. The design of the system is user-friendly. They used a CSS framework to have a basic structure of the system design, which includes grids and interactive user interface patterns.

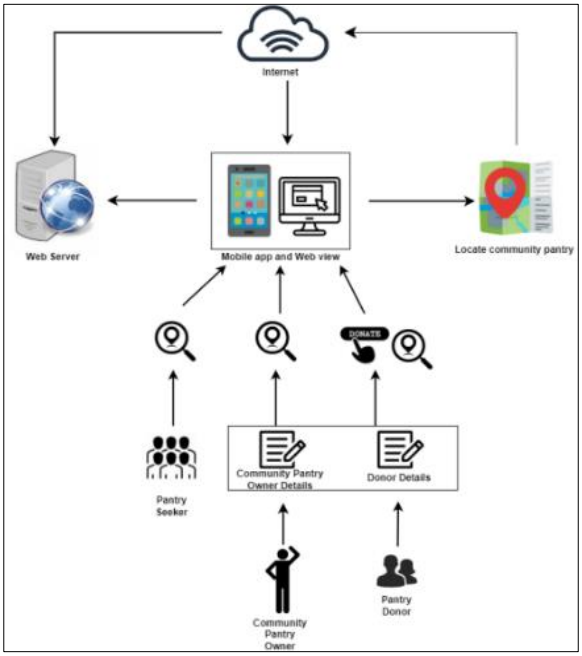

Figure 2. System architecture of the developed system

Figure 2 shows the system architecture that helps the researchers in defining the structure behavior, as well as additional views of the developed system. The system has a web view and mobile application for the citizens and it has three users, the community pantry owner who will create the post about their community pantry, the one who will donate and the pantry seeker who only just wants to find the different community pantry and they have the same interface. The community pantry owner can create post regarding to their pantry with the detailed information and they also need to create an account, the pantry donor will fill up the form for the detailed transaction of donation, their

name, phone number, schedule etc. and the pantry seeker or the one who only wants to find community pantry can view all the post about community pantries. With the help of GPS, they can also access all the community pantries on the map.

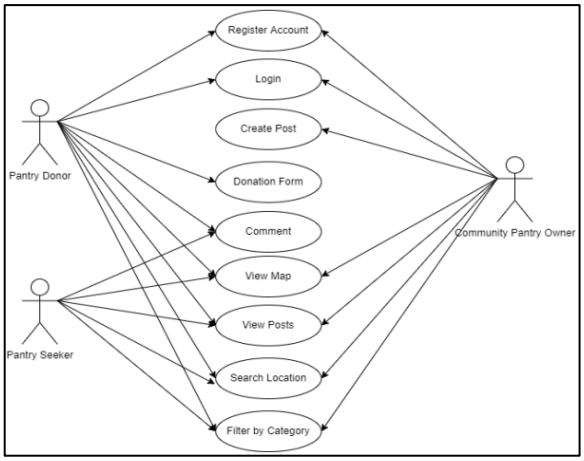

Figure 3. Use case diagram

Figure 3 shows the representation of the user's interaction to the system that has a mobile app and web system. It also shows the relationship of different users to the different use cases. The community pantry owner will register an account and login, then they can create a post regarding their community pantry and the pantry donor which they can donate based on what the community pantry needs. They can also access the different posts with the recommendation of the different community pantry owners and for the pantry seeker who only needs to know the different community pantries around their area or other places, they will just browse on it and they don't need to register an account. The pantry seeker can access the map that shows the locations of community pantries in the area represented by pins. By clicking on each pin, it shows the name of the pantries and the details

Building Phase

In this phase, the researcher started on building the codes by using the programming language that they chose. The developed system used appropriate commands and codes, because this phase took many iterations and rework for some defects that were encountered.

The develop application uses an ionic framework, it is created as a mobile application. Since the system is only available at android version, any IOS version such as iPhones won't work. On the other hand, you can't access the system without an internet connection.

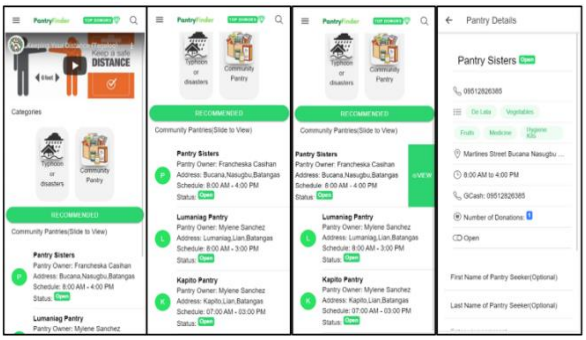

Figure 4. Pantry finder homepage

Figure 4 shows the homepage of the PantryFinder. In this interface the user can view the available pantries and the details of the pantry. The donor can also easily view the other details of the pantry and it also has the comment section for the donor and pantry seekers.

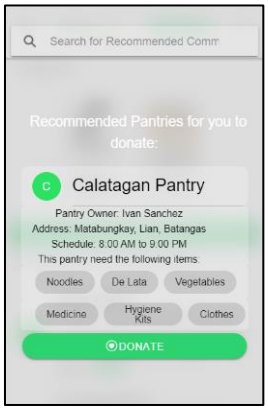

Figure 5. Recommendation page

Figure 5 shows the recommendation page for the PantryFinder. On this interface it has a donate button in every pantry, it is easy for the donor to donate in the pantry.

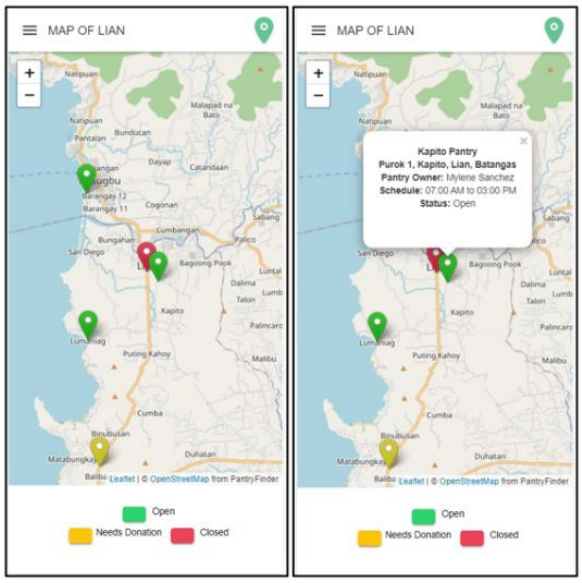

Figure 6. Pantry finder map

Figure 6 shows the map page where the user can see the different community pantries.

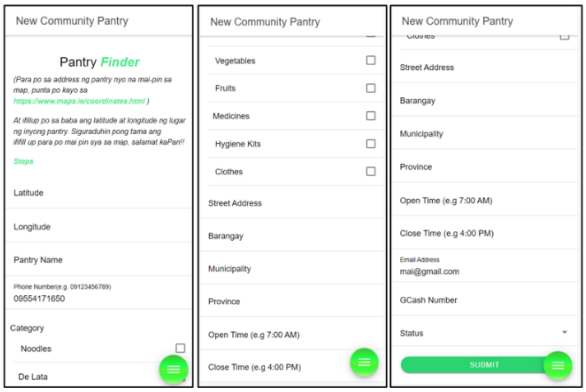

Figure 7. Add pantry page

Figure 7 shows the add pantry of the PantryFinder. On this interface it has a longitude and latitude to view your pantry on the map. It has the checkbox for the category of your pantry, the other information that will be needed to the created pantry. And also the submit button to successfully add the new pantry.

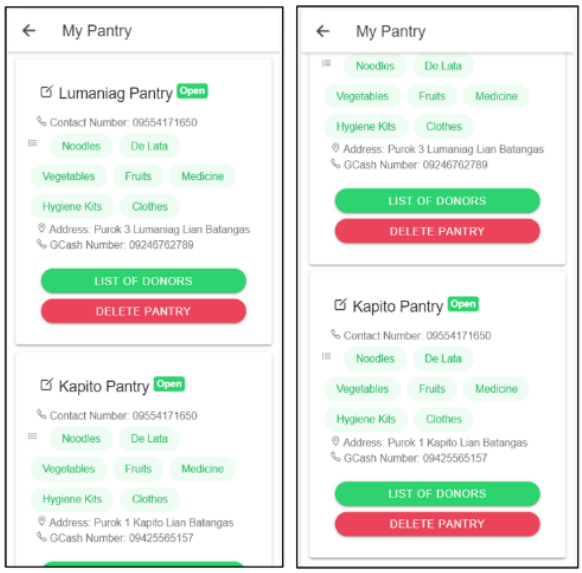

Figure 8. List of pantry

Figure 8 shows the list of pantries of the PantryFinder. In this interface it has a list of donors button and delete pantry button. And also, it has the edit button for the profile of the pantries.

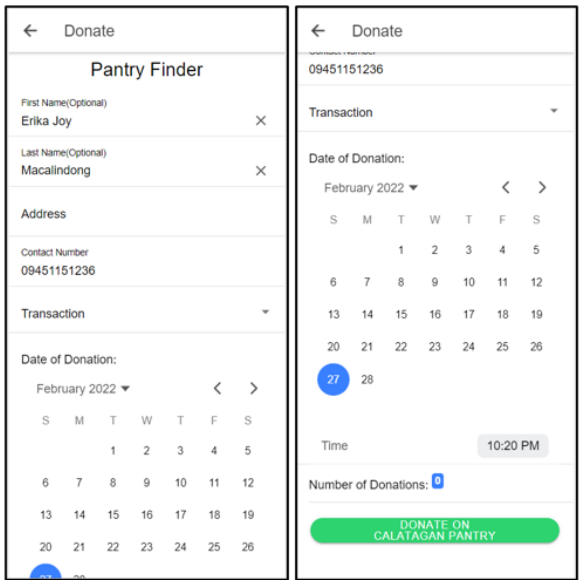

Figure 9. Donate page

Figure 9 shows the donation page and there's a field where the user will enter the information regarding to their donations. First ame and Last Name are optional, donors can delete it if they want to be anonymous. Donor will also input their address, type of transaction and the date they want to give their donations.

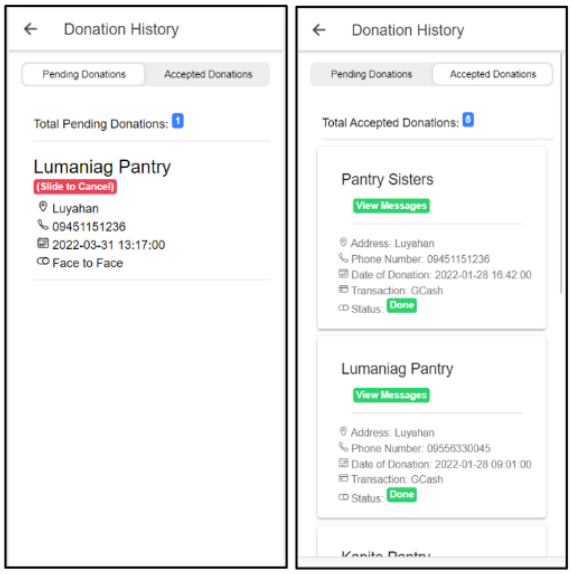

Figure 10. Donation history

Figure 10 shows the donation history of the PantryFinder. In this interface you can see the approved donations and you can monitor the status of the donations. It also has an interface that will show a message once it's approved by the owner

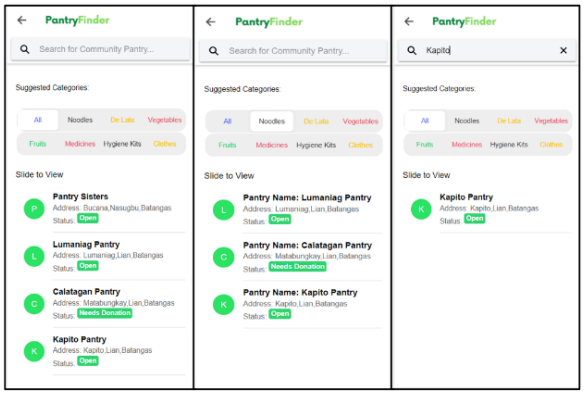

Figure 11. Search page

Figure 11 shows the search page of the PantryFinder. It has an interface that will show the suggested categories and it has the search bar. In the search bar you can type the names of the pantry or the names of the owner.

Testing Phase

In this phase, they run a test of the system to know if there's bugs. In acceptance testing by an end user, it will determine the validity of the system when used in a real situation.

Evaluation Phase

This phase evaluated all the processes and functions of the system. It determined the effectiveness of the program and its tools, and provided possible recommendations to improve the solution to fit best for the system and the needs of the program.

The researchers conducted the evaluation of the developed system to the 50 respondents including forty (40) pantry seekers of Lian, five (5) pantry owners and five (5) pantry donors. The researchers prepared videos on how to access the application based on their roles, so that respondents could view and know how to use it. They also provided three (3) sets of survey questionnaires by the use of google forms and they send the link of questionnaire and videos via Messenger to every respondent for them to watch and answer it.

In analyzing the gathered data, the researchers used the Likert Scale that is a 1–5-point scale that has a range of answer options to get the weighted mean and the verbal interpretation, as shown in Eq. (1)

$$Interval(i) = \frac{\text{HV}-\text{LV}}{\text{t}} \quad (1)$$

Where in: HV = Highest Value

LV = Lowest Value

t = Total Values

The pantry owner, pantry donor and pantry seekers are also asked to answer the level of acceptability of the developed system in terms of functional suitability, performance efficiency, usability and security.

Table 3. Guideline interval for level of acceptability

Scale	Mean Range	Descriptive Equivalent
5	4.21-5.00	Highly Acceptable
4	3.41-4.20	Moderately Acceptable
3	2.61-3.40	Acceptable
2	1.81-2.60	Slightly Unacceptable
1	1.00-1.80	Not Acceptable

Conceptual Framework

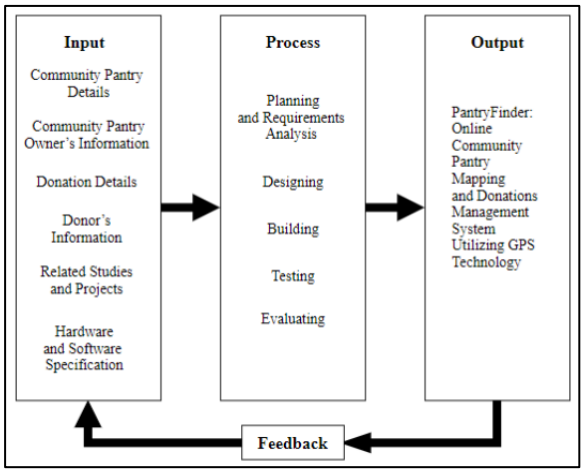

Figure 12. Conceptual framework

Figure 12 shows that the researcher must collect (a) community pantry details (b) community pantry owner's information (c) donation details (d) donor's information (e) related studies and projects and (f) the hardware and software specification. Upon collecting all the input, it will be processed by (a) planning and requirements analysis (b) designing (c) building (d) testing and (e) evaluating. After completing the process, the output will be PantryFinder: Online Community Pantry Mapping and Donations Management System Utilizing GPS Technology.

4. RESULTS AND DISCUSSION

This section presents the results of the evaluated data from respondents with reference to the objective of the study.

(a) Based on the evaluation, the problems were encountered in the management of community pantry:

Table 4. Problems encountered in the management of community pantry by the pantry owner

Problems	Frequency	Rank
Community pantry owners have difficulties in finding donors.	2	2
Donations of donors are not sustainable.	2	2
Many people arrived and there's insufficient items in that community pantry.	4	1
Community pantry owners have difficulties in knowing how to inform donors about the shortage of items.	1	3
Find a place or spot where the pantry can be easily accessed.	1	4
Looking for volunteers to repack the items/donations.	1	4
Seeking or encouraging in-kind donations.	3	2

Table 4 shows that "Many people arrived and there's insufficient items in that community pantry" got the highest rank and the researchers find that most of the pantry owners are having difficulties in managing the people who arrived and there's insufficient items in community pantries in the existing system.

Table 5. Problems encountered in the management of community pantry by the pantry donor

Problems	Frequency	Rank
Having difficulties in communicating with Pantry Owner	4	1
Having difficulties in finding a Community Pantry.	3	2
Looking for a pantry owner to make donations.	3	2
Not be able to identify the number or quantity of items to donate in a community pantry.	2	3
Repacking donations to give to the pantry owner.	1	4

Table 5 shows that "Having difficulties in communicating with Pantry Owners" got the highest rank and with this, the researchers find that the pantry donors are having difficulties interacting with the Pantry Owner because there's no direct communication.

Table 6. Problems encountered in the management of community pantry by the pantry seekers

Problems	Frequency	Rank
Having difficulties finding community pantries around their areas.	21	1
There is no exact location or area where the community pantry will be located.	20	2
Not being informed if the pantry they were going to is available at that time.	17	3
Not being informed if there is a new open pantry in their barangay.	16	4
They have no idea what available items they can find in the pantry.	14	5

Table 6 shows that "Having difficulties finding community pantries around their areas" got the highest rank and the researchers find that most of the pantry seekers are having difficulties in finding community pantries.

(b) After the requirements analysis phase, the modules for Pantry Owner, Pantry Donor and Pantry Seeker is considered for the developed system.

Table 7. Pantry owner's module

Module 1	Creating Account for Owner
Module 2	Login as Owner (Valid Owner's Account)
Module 3	Login as Owner (Invalid Owner's Account)
Module 4	Owner's Profile
Module 5	Map
Module 6	Add Pantry
Module 7	Owner's Pantry
Module 8	My Pantry
Module 9	List of Pantry
Module 10	Pantry's have Donations
Module 11	Owner's Account for Logout

Table 8. Pantry donor's module

Module 1	Creating Account for Donor
Module 2	Login as Donor (Valid Donor's Account)
Module 3	Login as Donor (Invalid Donor's Account)
Module 4	Donor's Profile
Module 5	Map
Module 6	Donor's Viewing for List of Pantry
Module 7	Donor's Donations
Module 8	Donor's View Community Pantry Details
Module 9	Donor's Donation History
Module 10	Donor's Account for Logout

Table 9. Pantry seeker's module

Pantry Seeker's Module	
Module 1	Viewing the Pantry Details
Module 2	Map

Modules represent each interface of the Pantry Owner, Pantry Donor and Pantry Seeker in the developed system. It includes modules for login, viewing community pantries, viewing maps, adding pantries and modules for donating.

(c) Through evaluation, the developed system used common hardware and software requirements for the application to run. Mobile phones that have at least 4GB of RAM, a version of android 8 to 10 and a laptop that has any version for the user admin.

Table 10. Software and hardware components

Particulars	Specification
Processor	2 GHz
Memory	4 GB
Storage	32 GB
Mobile Phones	Android Version
Phone OS	Jelly Bean up to Nougat
Web Browser	Google Chrome

(c) The test plan is used for the technical evaluation of the developed system and it shows to verify if the application is working properly.

Table 11. Level of acceptability of the developed system of the pantry owners, pantry donors and pantry seekers in terms of functional suitability

Parameters	Weighted Mean	Verbal Interpretation
The developed system provides correct results and details of the Community Pantry.	4.85	Highly Acceptable
The developed system provides the correct location and details of Community Pantry on the map	4.72	Highly Acceptable
The developed system shows the accurate number of donations and donors	4.68	Highly Acceptable
The developed system recommends Community Pantry that needs donations.	4.82	Highly Acceptable
Overall Weighted Mean	**4.77**	**Highly Acceptable**

Table 11 shows the level of acceptability in terms of functional suitability evaluation of the respondents. It reveals that the developed system is Highly Acceptable and convenient to use.

Table 12. Level of acceptability of the developed system of the pantry owners, pantry donors and pantry seekers in terms of performance efficiency

Parameters	Weighted Mean	Verbal Interpretation
The developed system helps the different users to do the action without difficulty in terms of creating a community pantry, donation and finding community pantry.	4.61	Highly Acceptable
The developed system helps the user to do action with less time and effort.	4.61	Highly Acceptable
Overall Weighted Mean	**4.61**	**Highly Acceptable**

Table 12 shows the level of acceptability in terms of performance efficiency evaluation of the respondents. It reveals that the developed system is Highly Acceptable and developed system helps the user to make the work easier in terms of giving donations, finding a pantry owner and searching for available pantries.

Table 13. Level of acceptability of the developed system of the pantry owners, pantry donors and pantry seekers in terms of usability

Parameters	Weighted Mean	Verbal Interpretation
The developed system presents the information of different community pantries in a clear and understandable way.	4.86	Highly Acceptable
The developed system uses tools that's easy to navigate and appropriate to use.	4.86	Highly Acceptable
The developed system supports the needs of the intended user in terms of finding the exact location of the community pantry.	4.8	Highly Acceptable
The developed system has an interface that is pleasing and satisfying for the user	4.79	Highly Acceptable
The developed system has instructions for users to avoid entering wrong information.	4.78	Highly Acceptable
Overall Weighted Mean	**4.82**	**Highly Acceptable**

Table 13 shows the level of acceptability in terms of usability evaluation of the respondents. It reveals that the developed system is Highly Acceptable and the developed system is extremely easy to use and comprehend, and it can meet the needs of the intended user and it has instructions to avoid misuse with regards to the developed system.

Table 14. Level of acceptability of the developed system of the pantry owners, pantry donors and pantry seekers in terms of security

Parameters	Weighted Mean	Verbal Interpretation
The developed system only allows the authorized person to access the needed information	4.62	Highly Acceptable
The developed system ensures safety of information and identity of the users	4.55	Highly Acceptable
The develop system has terms and conditions before creating an account for the security of their private information.	4.84	Highly Acceptable
Overall Weighted Mean	**4.67**	**Highly Acceptable**

Table 14 shows the level of acceptability in terms of security evaluation of the respondents. It reveals that the developed system is Highly Acceptable.

The users are convinced that the developed system will ensure the safety of their identity when they are creating an account, because they are the only one who can access their account.

5. CONCLUSIONS

1. Based on the findings, most of the pantry owners, pantry donors and pantry seekers were having difficulty on managing community pantries, the problems they encountered were the insufficient items, communicating with pantry owners, and difficulties in finding pantries. The developed system has a solution to the existing problems, provide the details regarding the available items, the contact number and address and by using the pantry map to easily find the available pantry in the barangay. This helped the researchers to create the PantryFinder so that they can solve the problems of the pantry owners, pantry donors and pantry seekers.

2. Based on the test case, each module was suitable for each scenario and has the correct result for the pre conditions you have entered. The researchers concluded that by entering the correct information, all the users can view all the details of the community pantry and pantry owners and donors can login in their own account to access every module.

3. The system is convenient to use, when the hardware and software you employ are appropriate for the system you're developing. And also to provide the functionality of the web system and the applications.

4. Based on the findings, the features of the developed system are functioning well in every module of the developed system according to the evaluation of the pantry owner, pantry donor and pantry seeker.

5. The evaluation revealed that the developed system is Highly Acceptable by the pantry owners, pantry donors and pantry seekers, and was proven to be more functional, efficient and secure. The researchers concluded that the features of the developed system will be a big help for them when it comes to sending donations, finding for the pantry owners or donors, and viewing all the information of the community pantry.

6. RECOMMENDATION

1. It is suggested for the integration of an advanced mapping system where it will show the distance of the community pantry from your location.

2. The researchers recommend making an inventory system of the essential items on a particular community pantry for the pantry seeker to inform what the exact number of items are still available.

3. The system should have a chat feature wherein the pantry owner and donor can communicate with each other when it comes to donation and receiving donations.

4. The researchers recommend a downloadable pdf file of certificates for the top donors in the application as a reward for their donations

5. The system must generate a control number for pantry seekers in order to line up properly and be able to control how many people would go to that pantry. And also, it must show the number of people that the pantry helped for the donors to see it.

6. For future researchers, the application should also have an iOS application so that everyone can use it. Also, recommends enhancing the user interface to make it more user friendly and appealing to the users.

7. REFERENCES

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Clickboat: Boat repair and rental Management System

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RATIONALE

The nature of boat repair and rental counting are time consuming and requires workforce and more effort to do. Manual process may also be hard to be done on a regular basis. By using the proposed project, boat repair and rental transaction processes will be automated and will be easy to manage on a daily and regular basis. The yacht owner, yacht charter, tourist and boat owners will be able to save time and effort such as notifications and updating transaction status.

OBJECTIVES

The general objective of the study is to develop a web/mobile application that can be used by yacht owner, yacht charter, tourist and boat owners. To be able to understand the main objective, the specific objectives were drawn: (1) Define problems encountered in: a. Yacht repair scheduling ,b.Yacht repair management, c. tourist iterinary ,d. boat scheduling;(2) Define modules of the system developed to address the problem;(3) Define the tools and technology used to develop the solution; (4) the level of acceptance of the respondents in the developed system in terms of: a. user friendliness , functionality ; (5) Issues encountered during the initial deployment of the system;(6) Features recommended to improve the system.

METHODOLOGY

To be able to develop the final output, the study followed the processes involved in the Agile Development. It includes all the processes that is necessary to finish the study which are: (1)Plan , (2)Design, (3)Develop (4)Test, (5) Release , (6) Feedback. The data needed to conduct the study were analysed and validated based from the system requirements of the project.

RESULTS

The system's evaluation resulted with the mean range of 4 above with the equivalent of high acceptability rating for both user friendliness and functionality. The developed modules are for tourist, yacht charter, yacht owner and boat owner where they can communicate and automate some of the process that involves notification and report generation.

CONCLUSIONS

Mobile and web application is a good way to cater marina based services such as boat repair and rental. Different available API and libraries are very useful in the development because it speeds up the process. The ability to test different environment of the system is a must especially technology is evolving.

KEYWORDS:*Yacht charter, Tourist Itinerary, Web application, Boat Scheduling, Yacht repair scheduling*.

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1. INTRODUCTION

The boating and yacht industry has been one of the major components in terms of marina tourism, Boat rental and repair has been vital yacht charter maybe not a typical type of business even though owning yacht is pretty expensive clients are still availing and finding services for boat repairs, same as the boat rental industry which may comprises tourism such as boat ride and island hopping , in addition to that Philippines is an archipelago that makes it a good spot for marina services such as marina tourism where the Tourist can book boat ride on Boat owners for tourists spot islands such as Fortune Island and Twin Island on Nasugbu,Batangas. The Country is also a hub for numerous number of Yacht charter that offers repairs on either boat or ships. Even though in some countries like US there are unemployment rate increase that makes the market on boating to decline but still the industry revenue manages to come back on the market with some constraints on the line such as smaller market but few competitors around due to unemployment of the crews.

Papaya yacht charter is one of the local yacht charter that offers this kind of service on the municipality of Nasugbu, they often use paper and flat database like excel on recording transaction, because of this report generating can take time to do and another is the handling of transactions is hard when it is open season or occasion of yacht repairs and maintenance transactions record are hard to keep because it is on the form of paper records that is probable of it to lost and will not be able to retrieve it. Another is the Boat owners have difficulty in terms of managing the transactions in terms of different tourist, tourist may also encounter difficulties of looking boat owners who has the right size of boat to rent. Barangay Papaya has been a tourist spot on the past years since 2015, manual transaction has been the constraint they have been facing that makes it difficult for the boat owners to cater guests.

The purpose of this research titled “Click Boat : Boat Repair and Rental Management System” is to improve and the business process on the industry of Boat repair and rental with the use of technologies and develop a system to serve both sides of yacht owners,manager, boat owner and tourists to help improve the aspects of the business such as report generating, tracking and management of the information needed to became a functional and efficient service provider. .

2. OBJECTIVES

The research aim to develop ClickBoat: Boat Repair and Rental Management System.
Specifically it sought the following:

1. What are the common problems encountered by the respondents in terms of:
   - 1.1 Tourist’s Itinerary
   - 1.2 Boat Scheduling
   - 1.3 Yacht Repair Scheduling
   - 1.4 Yacht Repair Management
2. What are the modules of the system to be develop to address the common problems encountered by the respondents?
3. What are the tools and technology did the Researchers use to develop the solution to the problem?
4. What is the level of acceptance of the respondents in the developed system “Click Boat: Boat Repair and Rental Management System” in terms of?
   - 4.1 User Friendliness; and
   - 4.2 Functionality
5. What Issues do the researchers encounter during the initial deployment of the “Click Boat: Boat Repair and Rental Management System”?
6. What are the features recommended by the respondents to improve the functionality of the system?

3. LITERATURES REVIEW

In different cases business automation is needed on a business as it grows from medium to large, Yacht repair servicing businesses and boat rentals start to use technologies to improve their performance and bring the customer to a better experience on giving the service, West Marine states that it uses its e-commerce website to test markets for new products in a cost-effective manner. [1]. the web became a famous platform over the past years; benefits that it offers are mutually equal to the side of the businesses and its customers. Web applications became famous because it does not require the user to install any additional software to access the service it is just one click away on their browser no matter what device (mobile, laptops or desktops) they are using. China yacht industry will maintain a market growth rate of no less than 30% per year, China is expected to seize the opportunity to welcome the future transfer of the yacht industry and become a new focus of the world's yacht industry[2].With the increasing numbers of the yacht, the needs for the existing yacht to repair became another likely service needed by the owners which later became the part of the YSC, with also the increasing numbers of repair to be done yacht charters start to employ repairmen to offer yacht repair and maintenance services.

On the other hand findings of IBIS World say that “The revenue of repairs and maintenance of the yacht is expected to rise over the years till the year 2021[3]”. The two studies both say that the market on yacht servicing is unsaturated; owners of yacht are ready to find charters that offer yacht repair and ready to cross the borders to avail the service. Hurley states numbers and statistics regarding the needs for the yacht servicing and systems that already exist in the market that is also used by different yacht charter [4] , which also corresponds to the research during the year 2018 have conducted in china that the need and demand for e-platforms for yacht related business are high[5].Progressive web apps are now on the demand and are being popularly used nowadays because of their similarities on mobile applications, WANG, ZHENG, and CHEN findings on the research entitled Analysis of the Capital Composition for Yacht Industry in China tell that the rapid development of the yacht industry bring different problems such as difficulty to adapt the changes happening on the current market, both of Wang and his fellow researcher in China presents the existing problems such as strengthening the market with the use of technology such as cloud and the web itself to cater a wide array of services[6].

Researchers from the California surveyed 130 people who had downloaded the RideScout app to evaluate attitudes and perceptions toward mobile apps, travel behaviour, and modal shift [8]. Mobile apps and web app became a trendy way to make it easy to bring marina services on the hand of the clients and service provider, although some issues has been found such as security and privacy. To cope up with the business, service management platforms are created to deal with untidy management of transactions and data. While on the other hand adding new features to make it easy for the future end user of the system are continued to discover. Mobile apps or web apps the key here is the accessibility, the ability of the systems itself to adjust to the environment of the platform it is being access. The technology and social networking revolution in America is transforming every area of American life, including transportation. The advent of a range of new technology-enabled transportation services that take advantage of mobile communications technology has been one of the most significant shifts. Americans will have new mobility options because to technological advancements and social networking technologies. New, technology-enabled transportation services might eliminate the need for many of these services by providing Americans with more mobility options and better methods to explore these options owning a personal car is becoming more common among Americans, resulting in a considerable drop in vehicle travel.

This research wants to imply the benefits of using a Service-oriented Information system on the Boat Repair and Rental that may deal with yacht charter, local Bangka owner and tourists with the use of developmental research strategy and instrumentation such as User acceptance tests, besides, that this research also wants to gather recommendations for future improvements of the research, another goal of this Research is to develop a service oriented software that can solve problems encountered in a Boat repair and rental in terms of scheduling and management of services such as status updates of repair in email and easy report generation .Boat reservation and integrating geolocation API and features.

4. MATERIALS AND METHODS

The researchers will use the descriptive - developmental method of research in this study. Descriptive research was used to obtain information concerning the current status of the phenomena to describe "what exists" with respect to variables or conditions in a situation, as these enable statistical analysis to be made while developmental because of the proposed Service-oriented system that is to be develop. After the development phase, the Service-Oriented system will be evaluated by selected respondents. This method of research applied prominently in this study because it presented the problems the company is facing in terms of management, scheduling and report generation.

To determine the effectiveness of the software, a non-randomized type of sampling is applied to increase the accuracy of results, the chosen respondents in the research are those who have knowledge in the yacht cluster servicing and those who are in the target market of the business specified.

In developing the said service-oriented platform the researchers will use the Agile model as the framework in developing and finishing the project. This is development method is popularly used in development a system because it covers the loop holes the waterfall has such as difficulty in incremental changes.

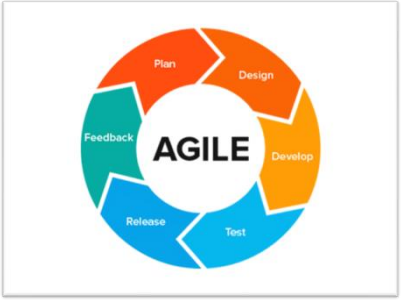

Figure 3.1: Agile Model

Figure above shows the Agile model. Agile Methodology is a people-focused, results-focused approach to software development that respects our rapidly changing world. It's centred on adaptive planning, self-organization, and short delivery times. It's flexible, fast, and aims for continuous improvements in quality [9].This method suits to the system that will be developed because this method can go back and forth to the previous and succeeding phases of the system development.

Plan

This phase is important because it will serve as the foundation of the development of the system and with this, the Researchers will do benchmarking and conduct a preliminary investigation on the existing business process of the stakeholder or client they are dealing with. In this phase also defines the scope of the system and system requirements.

Design

The process of defining the components, modules, interfaces, and data for a system to satisfy specified requirements. After the information has been gathered the Researchers will deal with the designing phase this phase includes the possible system architecture and the software tool needed to develop the said software.

Develop

This phase is more on the technical side of the process in the system, the Researchers will deal with programming and defining how to automate the business process in terms of translating it into a system requirement, in this phase also may include a prototype of the system and versions of it upon revision or reworking.

Test

After the development has been finished the output will undergo the testing process, different series of testing will be done to provide back job on the side of the developers. The Researchers will also deal on concluding what must be and what must not appear on the holistic features of the system.

Release

A release phase deals with some sort of experimental procedures even though the system passed the testing phase, it will still need to undergo on the experimentation because the environment of testing and operational is different, this phase also wants to know the feedback of the future end users of the system.

Feedback

In feedback this wants to know what to add, remove and remain on the features present on the system, the feedback also want to get the insights of the end users and deal with different issues such as data anomaly and Usability rating of the system, so that the output is ready to be deployed on the actual operational environment of the organization. after identifying the highest possible count of fish.

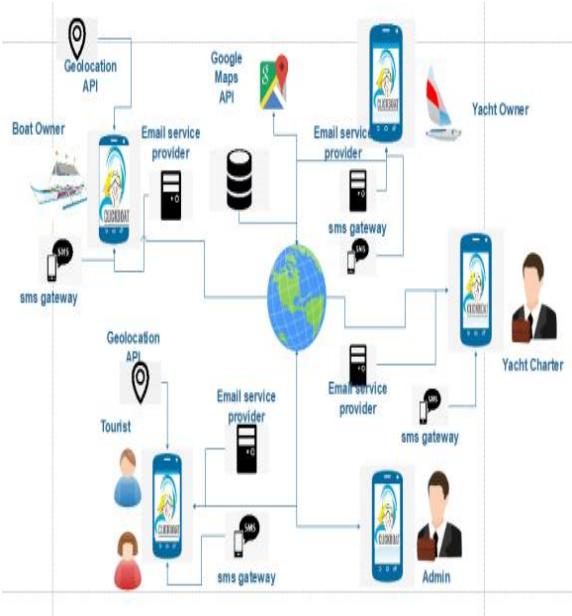

Figure 2. System architecture of the Developed Application

Table 1. Mobile App Requirements

Hardware	Technical Requirement	Purpose
Android	Android devices with Android 7+	Operating system
RAM	2gb above	Memory
ROM(Read Only Memory)	16 gb above ROM	Memory

5. RESULTS AND DISCUSSION

Problems encountered by the respondents on the existing system

Formula for getting the percentage:

Y = P% * X

Table 2. Tourist itinerary

Problem	Frequency	Rank
Finding a good boat driver for the boat ride service is difficult.	39	1st
Finding a good destination to go with is hard.	35	2nd
Finding a destination using Google search alone is difficult.	25	3rd
Searching for comments and feedbacks from other tourists is difficult.	17	4th
Others ; • price are not cascaded on boat rides	3	5th

As shown in the table above the major problem faced by the respondents in terms of tourist itinerary is "*Finding a good boat driver for the boat ride service*" which is answered by the 39 (97.5) of the respondents it can be assumed that it is an untidy job or there is no actual platform that deals with this type of business process same goes by the difficulty in "*Finding a good destination*" which is answered by 35 (87.5) of the respondents while the problem on "*Searching for comments and feedbacks from other tourists*" is answered by the 17 (42.5) respondents and using only Google maps as a platform gets 25 (62.5) answers from the respondents which can be assumed that the lack of platforms for tourist itinerary is a problem for tourists.

Table 3. Boat scheduling

Problem	Frequency	Rank
Deal with bulk transactions especially during peak seasons.	20	1st
Monitoring the good service provided by the boat owner to tourist is not recognized.	19	2nd
Managing boat reservation and scheduling transaction in manual manner.	17	3rd
Others	0	0

The respondents were asked what are the problems they have encountered in scheduling of the boat ride service for the tourist, the results as show on the table that 20 (100%) of the respondents have experienced "*difficulty in dealing with bulk transactions especially during peak season*"; while 19 (95%) of the respondents have find it difficult on "*Monitoring the good service provided by the boat owner to tourist*" and 17 (85%) of them have find it difficult in terms of "*Managing boat reservation and scheduling transaction in manual manner*".

Table 4. Yacht Repair Scheduling

Problem	Frequency	Rank
Yacht repair appointment in manual manner is difficult.	10	1st
Yacht repair appointment is challenging due to service fee are not cascaded.	10	1st
Some repair updates of yacht is not timely	8	2nd
Others	0	0

The respondents were asked to indicate what are their problems on yacht repair scheduling, 10 (100%) of the respondents answered "*difficulty in the manual process*"; 8 (80%) of the respondents have answered "*repair status update is not timely*" and also 10 (100%) have said that the "*repair scheduling is*

challenging due to the reason of service fee are not cascaded".

Table 5 . Yacht Repair management

Problems encountered	Rank
Using Log books and Record books to create a report is difficult.	1st
Difficulty to create sales invoice in a manual manner.	2nd
It takes so much time in preparing reports like annual reports, sales reports and invoice lists of finished transaction.	3rd
Difficulty in syncing information from different sources	4th

The table above shows the problems that the yacht charter have experienced using the manual system, Renato F. Lisud the manager of the Papaya Yacht Charter and Services Inc. said that although there is a system they are using in the company they find it hard to use it because it needed expertise in terms of accounting and proficiency in using flat database management systems such as Microsoft excel, one of the hindrance he also take note "the business lacks automation different clients are wanting to have the possible and easiest way to access the yacht repair services". As the researcher have demonstrate the technology solution to the Charter manager he have seen how the notification features of the system work on both Email and SMS he wants to have those kind of automation around their business saying that he wanted to implement it on the organization.

Modules of the System developed to address the common problems encountered by the respondents

Table 6. Modules of System developed to address the problem

Module name	**Access and Capability**
Tourist	• List of Destinations with current weather status, ratings and feedbacks. • List of Boat owner for boat reservation selection. • Geolocation View of Destination and current User location. • Providing rating for destinations and boat drivers • Crud functionalities on boat reservation such as Create, Cancel and update. • Receive Email and SMS notifications about transaction status. • Add proof of Down payment or Reservation fee for transaction assurance. • Password changing and retrieval. • User's profile
Boatdriver	• List of Boat reservation requests. • List of Previous Transactions and Tourist's rating. • Map view of the destination the tourist wanted. • Capability to update the Boat reservation's status. • Received SMS and Email notification about the status of boat reservation.
Boatdriver	• Verify the reservation fee or down payment of

	the tourist. • Password updating and retrieval facility. • View my profile and see data visualization of transactions
Yacht Owner	• Create, update and cancel a repair appointment. • View repair invoices. • Add repair feedback. • Receive invoice copy of repair in Email and SMS. • My profile • Password Retrieval and Updating.
Yacht charter	• List of repair appointments • Update Repair Status and generate invoices in PDF attached in the email notification to the clients. • Generate reports and view simple data visualization. • Calendar views of repairs. • View client's feedbacks. • My profile • Updating and Retrieval of Password. • Manage repairs and Services.

The table 8 shows the modules of the system the Researchers have developed to address the problems of the respondents shown in table 4, 5, 6 and 7 which describes the problem they encountered in using the manual or traditional business process. With these the Researchers have come up to the idea of creating modules that will automates the business process in terms of report generation and transaction awareness through system generated notifications that involved SMS and Email.

Tools and technology used to develop the solution to the problem

Table 7. Tools and technology used to develop the system

Tool or Technology used	**Type**	**Purpose**
Text editor tool	computer program	Edit plain text file, scripts and batch commands.
Xammp	cross-platform web server solution	Consisting mainly of the Apache HTTP Server, MariaDB database, and interpreters for scripts written in the PHP and Perl programming languages.
PHPmailer	Php class library file	PHPMailer is a code library to send emails safely and easily via PHP code from a web server.
DomPDF	Php class library file	HTML to PDF converter tool.
Bootstrap	CSS(Cascading Style Sheet) Framework	CSS Framework for developing responsive and mobile-first websites.
Filezilla	FTP client Software	Deploy Web app source codes such as Php,CSS and Javascript files on the Web Server.
Cpanel	Web-hosting control Panel	Manage Web apps configurations

		such as SSL certificates and other managerial task on website management,
Swing2app	Mobile App Builder	Create or Convert Web application into native mobile application.
Itextmo	SMS Gateway API	Send SMS notification in a specific mobile numbers.
OpenWeather	Weather API	Fetch weather updates and statuses.
Google Maps API	API	Fetch web mapping and geographic view of the specific location.
OpenStreetmaps	API	Create a free editable geographic database of the world.

The table above shows the technology and tools the Researchers used to develop the software in addressing the problem faced by the respondents, majority of these tools are used in creating progressive web apps and APIs are only used to give additional features on the system such as notification like Phpmailer and Itexmo does, while there are also present Geographic mapping tools used in the development such as Openstreetmaps and Google maps API.

Level of acceptance of the respondents in the developed system "Click Boat: Boat Repair and Rental Management System" in terms of?

5r=Number of Responses with 5

4r=Number of Responses with 4

3r=Number of Responses with 3

2r=Number of Responses with 2

1r=Number of Responses with 1

T=Total number of Respondents

Formula:

Mean= ((5r*5)+(4r*r)+(3r*3)+(2r*2)+(1r*1))/T

Mean2=((5r*25)+(4r*16)+(3r*9)+(2r*4)+(1r*1))/T

Standard Deviation=√ (Mean2-Mean)

Average= (∑mean)/number of parameters

Table 8. System Acceptability Rating In Terms Of User Friendliness

Parameters	Mean	Descriptive Equivalent
The system can adapt on different screen sizes.	4.68	Highly Acceptable
The UI design is learnable and easy to use/operate	4.58	Highly Acceptable
The system is easy to navigate and operate.	4.70	Highly Acceptable
The system provides different interactive views of information.	4.66	Highly Acceptable
The system provides a convenient way of password retrieval process.	4.58	Highly Acceptable
The system interface is aesthetically designed	4.66	Highly Acceptable

The table above shows the level of acceptance of the respondents in terms of the system's user-friendliness it shows a Standard deviation that is not below or greater than 4 which means that the data is not well dispersed or spread out. The mode is either 5 or 4 which tells moderate and high acceptability of the system's User friendliness. The traditional tourism industry is unable to follow with the development trend and with the emergence of modern personalized information services, and the service provided by the tourism e-commerce environment also important for improving the tourists' satisfaction and also the competitiveness [10].A different system has been on the trend to provide newer experience to the target market to avoid churns and decline of possible clients.

Table 9. System Acceptability Rating In Terms Of Functionality

Parameters	Mean	Descriptive Equivalent
The system provides accurate ratings, insights and data visualization	4.76	Highly Acceptable

The system improves the way the business process works.	4.63	Highly Acceptable
The system is capable of handling errors on some cases like invalid inputs such as dates and text strings.	4.63	Highly Acceptable
The system can aid the user to finish a specified task.	4.60	Highly Acceptable
The system can facilitate the accomplishment of specified tasks and functions.	4.62	Highly Acceptable

The table above shows the level of acceptance of the respondents in terms of the system's Functionality it shows a Standard deviation that is not below or greater than 4 which means that the data is not well dispersed or spread out. The mode is either 5 or 4 which tells moderate and high acceptability of the system's Functionality. While the incorporation of some earlier RAISA technologies such as the ATM may have radically changed the way the customer experiences [11].The concept of this information system is to automate the system even though the automation itself is not in the form of holistic automation but automating some of the most vital parts of the process such as analytics, notifications and other informatics related function that this system can bring on user and persuade to use newer systems.

Issues encountered during the deployment of the System

Table 10. Deployment Issues

Issue	Cause	Severity
DNS error	Web server Downtime	Major
MySQL error	Web server Downtime or TCP/IP socket not listening	Moderate
Web Mail Not working	email account has exceeded its quota IMAP/POP is not enabled.	Minor
Website suspension due to usage limits	Exceeded web server resources.	Minor
FTP service not working	Webserver Downtime	High
GPS not working	GPS service is not working on non-encrypted connection like HTTP (port 80)	Cosmetic
App not installed	Android version is not compatible to the app package.	Moderate
Cron Jobs Not working properly	Execution time exceeded. Incorrect permission.	Moderate
Website not safe alert	SSL certificate problem, DNS propagation in progress	Moderate

The following data on the table shows different issues that the Researcher have encountered during the deployment of the System that is according to the ISQTB (International Software Testing Qualifications Board), to sum it up, it can be seen can seen that a web based systems may encounter unexpected problems such as downtime due to multiple services and requests it is receiving across the network. In the 24/7 web industry, business owners can feel a constant need to push the latest and greatest software to the live site in order to remain competitive and ensure growth (John ,Mostashari and Mansouri,2011). It is important to always have a backup and select the best Web service package service as soon as possible because cheap services may bring drawbacks as the system scales up such as increasing volume of users and data that it process, In addition to that cross platform system such as combination of Native and mobile apps are a bit complicated due to the way they access the data and present it to the users.

System features recommended by the respondents

Table 13. Recommended features

Features	Frequency	Rank
Automated SMS and Email notification	67	1st
Payment Gateway Facility	62	2nd
Google Maps Direction	57	3rd
Simple analytics	39	4th

The implication of this finding is the respondents have suggests different features

that may improve the system for later purposes, to explain that the respondents wanted to automate almost the entire process on the business which involved notification features. The payment facility is the second majority feature they want to be added, with these result we can assume that the need of the users for automation for payment process is a must implemented feature on the system, On the Geographic features the respondents wanted to add the Google map direction services because this gives a better routes on the users while the analytics have the least number of recommendation it might be assumed that respondents have less experience or knowledge to the analytics itself.

6. CONCLUSIONS

From the above summary, the following can be concluded from this study:

- A manual process on boat repair and rental is difficult to deal especially as the data or information that the user is managing scales up especially on Boat rental where the transaction needed to process increases on peak seasons, tourists have a problem where destination to go and who else to contact to access the service. While on the other hand on yacht servicing clients that needed to avail repair services face challenges in creating a repair appointment due to price not being cascaded, paper-dependent processes, and lastly the repair management of Yacht charter is difficult because of different data sets from different flat file or paper sources that may involve logbooks, stand-alone system, and flat databases they are using, the need for automation can be seen on the problems they are facing that might resolved by the creation of a platform that can facilitate report generation and notification automation.
- The need for the development of a service-oriented platform that is intended to be used by tourists, yacht owners, boat drivers, and yacht charters where they can avail and provide services across the internet to adopt the new generation of age of paperless businesses.
- The internet itself plays a big role in the SDLC process it provides a readymade library for a fast phased development process that involves frameworks, API, libraries, and other existing systems that developers may benchmark to create innovative solutions for businesses and the community. Software severities are expected to arise on initial deployment due to incompatibility of legacy and newer systems
- Based on the findings of the acceptance test done, the study shows that it is difficult to measure the user-friendliness of the system due to the issues that users are not or does not have the same standards on Graphical interface and Cognitive understanding. The correct selection of colors, fonts and symbols must be considered to provide a user-friendly system. On the other hand, they find out that users wanted to have access to details about the information itself that may include rating and also they seek more on automation modules that may give them ease on accessing the service and the information.
- Good planning especially from a phase to phase basis such as holistic planning of deployment and procurement of resources must also discuss carefully. A series of tests from the different environments must be done first to assure the quality and assurance aspect of the system. The discrepancy between how the actual business environment and the unit testing environment is large. Software severity is a helpful tool to measure the defects and bugs of the software. In addition to that the because the deployment is one of the most vital processes of SDLC choosing the best package on technology to integrate is the best decision to do to avoid back jobs and troubles such as lack of bandwidth or performance of the system.
- The software evaluation of the study discovers that users wanted to have everything in touch of their hand; they wanted everything to be processed on just simple steps and clicks. Geographical and automated features are those features they want to have for the system improvement. There was evidence on the results that the system has a high acceptability rating on the respondents and room for its improvement is still possible especially if there is a budget for implementation and technology procurement.

7. RECOMMENDATIONS

The following are recommendations for future researches and implementation:

1.The developed system may improve its usage on a larger web hosting packaged.

2.Project budget is a very vital part of a study with these we can expand the quality of the system that we are going to develop.

3.The developers may add search box and sort the transactions based on its status and date creation.

4.Future developers may include features that can reach out other related people even though they are not the user of the system.

5.The future Researcher may use this paper as a reference related to business process automation related to Marina businesses. Accessing the system.

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